

Dated:

Question # 01)

Find:

Order, degree, linear, non-linear, PDE or ODE.

(i) $e^x dx + e^y dy = 0$
= Solution:

Order = 1
Degree = 1
Linear
ODE

(ii) $y = xy' + a[1 + (y')^2]$
= Solution:

Order = 1
Degree = 2
Non-linear
ODE

(iii) $y'' + x^2 y = 0$
= Solution:

Order = 2
Degree = 1
Linear
ODE

(iv) $y''' + 4y'' - 6y' + y = \cos x$
= Solution:

Order = 3
Degree = 1
Linear
ODE

(v) $y = xy' + x/y'$
= Solution:

Order = 1
Degree = 2
Non-linear
ODE

(vi) $u_{xx} + u_{yy} = 0$
= Solution:

Order = 2
Degree = 1
Linear
PDE

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(vii) $[1 + (y')^2]^{2/3} = c^2 (y'')^2$
= Solution:

Order = 2
Degree = 2
Non-linear
ODE

(viii) $(y')^2 = (y'' + y)^{3/2}$
= Solution:

Order = 2
Degree = 3
Non-linear
ODE

Question # 02

Obtain DE by Eliminating arbitrary constant

(i) $y = ae^x + bxe^x + cx^2e^x \rightarrow \text{eq (1)}$
= Solution: $u = e^x, v_0 = a + bx + cx^2$
Diff w.r.t x

$y' = ae^x + b(e^x + xe^x) + c(2xe^x + x^2e^x)$
 $u'v_0 + uv'_0 \leftarrow y' = \underbrace{e^x}_{u} [a + b(1+x) + c(2x + x^2)] \rightarrow e^x \text{ (common)}$

$y' = e^x [a + b + bx + 2cx + cx^2] \rightarrow \text{eq (2)}$

App. Product rule

$u'v_1 + uv'_1 \leftarrow y'' = e^x [a + b + bx + 2cx + cx^2] + e^x [b + 2c + 2cx]$

$y'' = e^x [a + 2b + 4cx + cx^2 + bx]$

$y'' = \underbrace{e^x}_{u} [a + 2b + \underbrace{(b+4c)}_{v_2} x + cx^2] \rightarrow \text{eq (3)}$

$u'v_2 + uv'_2 \leftarrow y''' = e^x [a + 2b + (b+4c)x + cx^2] + e^x [(b+4c) + 2cx]$

$y''' = e^x [a + 2b + 2c + (b+4c)x + cx^2 + b+4c+2cx]$

$y''' = e^x [a + 3b + 6c + (b+6c)x + cx^2] \rightarrow \text{eq (4)}$

$y''' - 3y'' + 3y' - y = 0 \text{ Ans.}$

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(ii)

$$y = e^x (A \cos 2x + B \sin 2x)$$

= Solution:

$$y = e^x (A \cos 2x + B \sin 2x) \rightarrow (1)$$

$$y' = u'v + uv'$$

$$y' = e^x (A \cos 2x + B \sin 2x) + e^x (-2A \sin 2x + 2B \cos 2x)$$

$$y' = e^x [A (\cos 2x - 2 \sin 2x) + B (\sin 2x + 2 \cos 2x)]$$

$$y'' = e^x [A (\cos 2x - 2 \sin 2x) + B (\sin 2x + 2 \cos 2x)] + e^x [\quad]$$

$$A (-2 \sin 2x - 4 \cos 2x) + B (2 \cos 2x - 4 \sin 2x)$$

→ simplified

$$y'' = e^x [A (-3 \cos 2x - 4 \sin 2x) + B (-3 \sin 2x + 4 \cos 2x)]$$

Find the pattern of y, y', y''

$$y = e^{mx} (A \cos nx + B \sin nx)$$

→ Because every part contain $e^x, \cos 2x, \sin 2x$

$$(D-m)^2 + n^2 = 0$$

$$\begin{aligned} (D-1)^2 + 2^2 &= 0 \\ D^2 - 2D + 5 &= 0 \\ D^2 - 2D + 1 + 4 &= 0 \end{aligned}$$

$$y'' - 2y' + 5y = 0$$

Ans

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Question #03

Solve for DE:

$$(i) (x^2 + y^2) dx + 2xy dy = 0$$

= Solution:

Homogeneous

$$\text{Let, } y = vx \Rightarrow dy = v dx + x dv$$

$$= (x^2 + v^2 x^2) dx + 2x^2 v (v dx + x dv) = 0$$

$$= x^2 (1 + v^2) dx + 2x^2 v^2 dx + 2x^3 v dv = 0$$

$$= x^2 [(1 + v^2) + 2v^2] dx + 2x^3 v dv = 0$$

$$= x^2 (1 + 3v^2) dx + 2x^3 v dv = 0$$

Divide both sides by x^3

$$= \frac{\cancel{x^2} (1 + 3v^2) dx}{\cancel{x^3}^1} + \frac{2\cancel{x^3} v dv}{\cancel{x^3}} = 0$$

$$= \frac{(1 + 3v^2) dx}{x} + 2v dv = 0$$

Separate var.

$$\frac{(1 + 3v^2)}{x} dx = -2v dv$$

$$2v dv = - \frac{(1 + 3v^2)}{x} dx$$

$$\frac{2v}{(1 + 3v^2)} dv = - \frac{dx}{x}$$

Dated:

Integrating both sides

$$= \int \frac{2v}{1+3v^2} \cdot dv = \int -\frac{dx}{x}$$

$$= \frac{1}{3} \ln |1+3v^2| = -\ln |x| + C$$

Multiply B.S by 3

$$= \ln |1+3v^2| = -3 \ln |x| + C$$

$$= \ln |1+3v^2| = \ln \left| \frac{C}{x^3} \right|$$

Removing log

$$= 1+3v^2 = C/x^3$$

Put $v = y/x$

$$= 1+3\left(\frac{y}{x}\right)^2 = C/x^3$$

= Multi. by x^2 on B.S

$$= x^2 + 3y^2 = C/x$$

$$= x^3 + 3xy^2 = C$$

$$= \boxed{C = x^3 + 3xy^2} \quad \text{Ans}$$

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$$(ii) \frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x) e^x \sec y \quad (\text{Solve for DE})$$

= Solution:

$$= \frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x) e^x \sec y \rightarrow \text{eq(1)}$$

multiply B.S by $(\cos y)$

$$= \cos y \cdot \frac{dy}{dx} - \frac{\cos y \cdot \tan y}{1+x} = (1+x) e^x \sec y \cdot \cos y$$

$$= \cos y \cdot \frac{dy}{dx} - \frac{\cancel{\cos y} \cdot \sin y / \cancel{\cos y}}{1+x} = (1+x) e^x \cdot \frac{1}{\cancel{\cos y}} \cdot \cancel{\cos y}$$

$$= \cos y \frac{dy}{dx} - \frac{\sin y}{1+x} = (1+x) e^x$$

Let,

$$u = \sin y$$

As,

$$\frac{du}{dx} = \cos y \frac{dy}{dx}$$

$$= \frac{du}{dx} - \frac{\sin y}{1+x} = (1+x) e^x$$

$$\text{As, } \sin y = u$$

$$= \frac{du}{dx} - \frac{u}{1+x} = (1+x) e^x$$

$$= \frac{du}{dx} - \frac{u}{1+x} = (1+x) e^x \rightarrow \text{eq(2)}$$

Find (I.F)

$$I.F = e^{\int -\frac{1}{1+x} \cdot dx} \Rightarrow e^{-\ln(1+x)} \Rightarrow \frac{1}{1+x}$$

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Multiply eq (2) by I.F

$$\begin{aligned} \cancel{I.F} &= \frac{1}{1+x} \cdot \frac{du}{dx} - \frac{u}{(1+x)^2} = \cancel{\frac{1}{1+x}} \cdot \cancel{(1+x)} e^x \\ &= \frac{1}{1+x} \cdot \frac{du}{du} - \frac{u}{(1+x)^2} = e^x \end{aligned}$$

As derivative is available in L.H.S

$$\begin{aligned} \frac{u}{1+x} &= \int e^x \cdot dx \\ &= u = (e^x + C)(1+x) \end{aligned}$$

Put $u = \sin y$

$$\begin{aligned} \sin y &= (1+x)(e^x + C) \\ \text{OR} \\ &= \end{aligned}$$

$$\boxed{y = \sin^{-1} (1+x)(e^x + C)}$$

Ans
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Question # 04

Initial value:

(i) $y'' - 4y' + 13y = 8 \sin 3x$ $[y(0)=1, y'(0)=2]$
= Solution:

$$y'' - 4y' + 13y \rightarrow \text{Homogeneous}$$

$$8 \sin 3x \rightarrow \text{Non } \neq$$

By using Quadratic formula:
L.H.S

$$\begin{cases} y'' - 4y' + 13y = 0 \\ \rightarrow x^2 - 4x + 13 = 0 \end{cases}$$

$$x = \frac{4 \pm \sqrt{(-4)^2 - 4(1)(13)}}{2} = \frac{4 \pm \sqrt{16 - 52}}{2}$$

$$x = \frac{4 \pm 6i}{2} = 2 \pm 3i$$

$$y = e^{2x} (C_0 \cos 3x + C_1 \sin 3x) \rightarrow \text{eq (1)}$$

R.H.S $\rightarrow$ Non $\neq$ homo.

$$y = A \cos 3x + B \sin 3x$$

$$y' = -3A \sin 3x + B \cos 3x$$

$$y'' = -9A \cos 3x - 9B \sin 3x$$

Put these
into L.H.S

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$$= y'' - 4y' + 13y = (-9A \cos 3x - 9B \sin 3x) - 4(-3A \sin 3x + 3B \cos 3x) + 13(A \cos 3x + B \sin 3x)$$

Simplification:

$$\text{For cos} \rightarrow -9A + 13A - 12B = 4A - 12B$$

$$\text{For sin} \rightarrow -9B + 13B + 12A = 4B + 12A$$

$$y'' - 4y' + 13y = (4A - 12B) \cos 3x + (12A + 4B) \sin 3x$$

Simplify for A and B

$$4A - 12B = 0 \Rightarrow A = 3B$$

$$12A + 4B = 8 \Rightarrow 12(3B) + 4B$$

$$36B + 4B \Rightarrow 40B = 8 \Rightarrow B = \frac{1}{5}$$

$$A = 3B \Rightarrow A = \frac{3}{5}$$

$$y = \frac{3}{5} \cos 3x + \frac{1}{5} \sin 3x$$

General solution:

$$y = y_h + y_p = e^{2x} (C_1 \cos 3x + C_2 \sin 3x) + \frac{3}{5} \cos 3x + \frac{1}{5} \sin 3x$$

$$y(0) = 1 \Rightarrow C_1 = \frac{2}{5}$$

$$y'(0) = 2 \Rightarrow C_2 = \frac{1}{5}$$

Final solution:

$$y = e^{2x} \left(\frac{2}{5} \cos 3x + \frac{1}{5} \sin 3x \right) + \frac{3}{5} \cos 3x + \frac{1}{5} \sin 3x$$

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⇒ Solve the DE:

$$(i) \quad \sin x \frac{dy}{dx} + 2y = \tan^3 \left(\frac{x}{2} \right)$$

= Solution:

Make it an standard form

$$y' + f(x)y = g(x)$$

Divide B.S by $\sin x$

$$\frac{dy}{dx} + \frac{2}{\sin x} \cdot y = \frac{\tan^3(x/2)}{\sin x} \rightarrow \text{eq (1)}$$

Find I.F

$$= I.F = e^{\int f(x) dx} = e^{\int \frac{2}{\sin x} \cdot dx} = e^{2 \int \csc x dx}$$

$$\text{As, } \int \csc x dx = \ln |\tan(x/2)|.$$

$$I.F = e^{2 \ln |\tan(x/2)|} = \left(\tan \left(\frac{x}{2} \right) \right)^2$$

Multiply eq (1) by I.F

$$\left(\tan \left(\frac{x}{2} \right) \right)^2 \cdot \frac{dy}{dx} + \frac{2 \tan^2(x/2)}{\sin x} y = \frac{\tan^5(x/2)}{\sin x}$$

R.H.S is exactly as $\frac{d}{dx}$ of L.H.S

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So,

$$y \cdot \tan^2\left(\frac{x}{2}\right) = \int \frac{\tan^5\left(\frac{x}{2}\right) \cdot dx}{\sin x} \rightarrow \text{eq(2)}$$

Let,

$$\tan\left(\frac{x}{2}\right) = t$$

After subs. $dt = \frac{1}{2} \sec^2\left(\frac{x}{2}\right) dx$

$$= \int \tan^x\left(\frac{x}{2}\right) \cdot dx$$

By using formula

$$\int \tan^3 u \cdot du = \frac{1}{2} \tan^2 u - \ln |\cos u| + C$$

Put $\left(\frac{x}{2}\right)$ back into original.

$$= \frac{1}{2} \tan^2\left(\frac{x}{2}\right) - \ln |\cos\left(\frac{x}{2}\right)| + C$$

For y : from eq(2)

$$y = \frac{\frac{1}{2} \tan^2\left(\frac{x}{2}\right) - \ln |\cos\left(\frac{x}{2}\right)| + C}{\tan^2\left(\frac{x}{2}\right)}$$

OR

=

$$\boxed{y = \frac{1}{2} - \frac{\ln |\cos\left(\frac{x}{2}\right)|}{\tan^2\left(\frac{x}{2}\right)} + \frac{C}{\tan^2\left(\frac{x}{2}\right)}}$$

Ans

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Solve for DE

$$(ii) \quad (x^3 - x) \frac{dy}{dx} - (3x^2 - 1)y = x^5 - 2x^3 + x$$

= Solution:

Divide B.S by $(x^3 - x)$

$$= \frac{dy}{dx} - \frac{(3x^2 - 1)}{x^3 - x} \cdot y = \frac{x^5 - 2x^3 + x}{x^3 - x}$$

$$= \frac{dy}{dx} - \frac{3x^2 - 1}{x(x^2 - 1)} \cdot y = \frac{x(x^4 - 2x^2 + 1)}{x(x^2 - 1)}$$

$$= \frac{dy}{dx} - \frac{3x^2 - 1}{x(x-1)(x+1)} \cdot y = \frac{x^1(x^2 - 1)^2}{x(x^2 - 1)}$$

$$= \frac{dy}{dx} - \frac{3x^2 - 1}{x(x^2 - 1)} \cdot y = x^2 - 1 \rightarrow \text{eq (1)}$$

Find I.F

$$I.F = e^{\int -\frac{3x^2 - 1}{x(x^2 - 1)} \cdot dx} = e^{\int -\left(\frac{3x^2 - 1}{x(x-1)(x+1)}\right) dx}$$

Use Partial Fraction:

$$= \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$$

$$3x^2 - 1 = A(x-1)(x+1) + Bx(x+1) + Cx(x-1)$$

For A, B, C
 $\downarrow \quad \downarrow \quad \downarrow$

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$$I.F = e^{-\left(\frac{1}{x} + \frac{1}{x-1} + \frac{1}{x+1}\right) \cdot dx}$$

$$I.F = e^{-(\ln|x| + \ln|x-1| + \ln|x+1|)}$$

$$I.F = \frac{1}{x(x-1)(x+1)}$$

Multiply eq.(1) by I.F

$$= \frac{1}{x(x-1)(x+1)} \cdot \frac{dy}{dx} - \frac{3x^2-1}{x(x^2-1)} \cdot \frac{1}{x(x-1)(x+1)} y =$$

L.H.S is simplified to derivative $\rightarrow$ R.H.S

$$\frac{x^2-1}{x(x-1)(x+1)}$$

$$\frac{x^2-1}{x(x-1)(x+1)}$$

Use Partial Fraction

$$\frac{x^2-1}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$$

$$x^2-1 = A(x-1)(x+1) + Bx(x+1) + Cx(x-1)$$

$$\text{For } A, B, C \Rightarrow A=1, B=0, C=0$$

$$\frac{x^2-1}{x(x-1)(x+1)} = \frac{1}{x}$$

$$\text{Integrate} \rightarrow \frac{d}{dx} \left[\frac{y}{x(x-1)(x+1)} \right] = \frac{1}{x}$$

$$= \frac{y}{x(x-1)(x+1)} = \ln|x| + C \quad \text{And}$$
$$y = (x^3-x)(\ln|x| + C)$$

Ans.